



National University
Polyclinics

MANAGEMENT OF CHRONIC CORONARY SYNDROME

Key Family Physician: Dr Kwan Yew Seng

Specialist Advisor: Dr Lim Toon Wei (Senior Consultant, Department of Cardiology, National University Heart Centre, Singapore)

Acknowledgement for their inputs: NUHSP: Mr Marvin Sim
Nursing: APN Liau Wei Fong

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INTRODUCTION

Chronic Coronary Syndrome (CCS) is also known as chronic or stable ischaemic heart disease or stable coronary artery disease.

It refers to clinical presentations that arise due to structural or functional alterations related to chronic diseases of the coronary arteries or microcirculation.

CCS encompasses six different clinical presentations:

1. Stable angina with suspected CAD $\pm$ dyspnoea*
2. Stabilised symptom(s) after ACS or revascularisation*
3. Asymptomatic CAD*
4. New-onset HF with suspected CAD
5. Vasospastic angina
6. Microvascular angina

**Clinical presentations covered in this CPG*

MANAGEMENT

The primary goals of CCS management are to:

- Reduce the incidence of first acute myocardial infarction in patients with screening-detected CAD
- Alleviate symptoms
- Reduce the recurrence of myocardial infarctions
- Prevent the onset of complications such as heart failure and atrial fibrillation
- Improve quality of life

This is achieved through a combination of pharmacological and non-pharmacological treatment, including lifestyle interventions.

Pharmacological Treatment

(A) Anti-Platelet Therapy

- Use long-term low-dose aspirin monotherapy for secondary prevention of cardiovascular events
- Long-term clopidogrel monotherapy can be considered as an alternative to aspirin
 - In patients with known CYP2C19 variants (Decrease / Poor Metaboliser), use of Clopidogrel is not recommended

- Low dose aspirin is associated with minimal bleeding risk. For patients at high risk* of gastrointestinal bleeding, gastric protection with proton pump inhibitors may be required.
- Use a PPI if gastric protection is required for patients on clopidogrel and ticagrelor#.
- Dual antiplatelet therapy (DAPT), combining aspirin with clopidogrel or ticagrelor, is indicated for patients after PCI due to its superior efficacy in preventing thrombosis compared to aspirin monotherapy, despite an increased bleeding risk. The optimal duration of DAPT is influenced by factors such as procedure-related considerations and the number of stents or stent sites. Therefore, DAPT duration for patients with CCS following PCI is typically determined by the cardiologist involved in the patient's care and communicated to the primary care physician.

** High risk: Patients with a history of ulcer complications, ulcer disease (non-bleeding), gastrointestinal bleeding and those with two or more of the following risk factors for gastrointestinal complications - age 60 years or more, corticosteroid use, dyspepsia or gastroesophageal reflux disease symptoms*

Although omeprazole and esomeprazole reduce exposure to clopidogrel's active metabolite, the effects of combination of omeprazole /clopidogrel on the risk of ischemic events or stent thrombosis have not been demonstrated. Discuss with cardiologist through VPC if necessary.

(B) Medications for Prevention of Angina

(1) Beta blockers and calcium channel blockers (CCB) (non-dihydropyridine and dihydropyridine)

- A beta blocker is typically used first line unless contraindicated
- Use a CCB when beta blockers are contraindicated or cause unacceptable side effects
- A combination of a dihydropyridine CCB and a beta blocker may be used if initial treatment with beta blocker is unsuccessful
- Avoid concurrent use of non-dihydropyridine CCB and beta blocker due to risk of heart block and bradycardia

(2) Additional options: long-acting nitrates, ivabradine (Not available in NUP), ranolazine (Not available in NUP) and trimetazidine

- Typically added to a beta blocker and/or a CCB if additional anti-anginal therapy is required
- Avoid concurrent use of nitrates with phosphodiesterase-5 (PDE-5) inhibitors (e.g. sildenafil) due to risk of severe hypotension
- Ivabradine should not be combined with non-dihydropyridine CCB
- When using long-acting nitrates, follow the recommended dosing instructions to minimise the risk of nitrate tolerance (refer to individual product information for more specific instructions)

(C) Management of Co-Morbid Conditions

Managing dyslipidaemia, hypertension, diabetes mellitus (DM), chronic kidney disease (CKD) and chronic heart failure (CHF) is crucial to reduce overall CV risk and complications.

(1) Patients with T2DM (Refer to [NUP Diabetes Mellitus CPG](#) for further information)

- Consider adding a SGLT2 inhibitor or GLP-1 RA with proven CV benefits, regardless of baseline or target HbA1c level, to reduce the risk of adverse cardiorenal outcomes
- Target HbA1c $\leq 7\%$; less stringent HbA1c target (e.g. $\leq 8\%$) may be appropriate in some patients (e.g. frail, older patients; patients with short life expectancy)

(2) Patients with hypertension (Refer to [NUP Hypertension CPG](#) for further information)

- Use an ACE inhibitor, ARB or calcium channel blocker as first line antihypertensive medications
- Consider thiazide/thiazide-like diuretics as alternative first line if indicated
- Target BP $< 130/80$ mmHg, with target adjusted based on individual patient circumstances. For example, less stringent targets (e.g. $140/90$ mmHg) can be considered in older patients (≥ 85 years of age) or patients with pre-treatment symptomatic orthostatic hypotension.

(3) Patients with dyslipidaemia (Refer to [NUP Lipids CPG](#) for further information)

- Treat with maximally tolerated dose of statin, with or without ezetimibe
- The addition of PCSK9i may be needed in some patients with high baseline LDL-C or requiring stringent LDL-C targets
- Target LDL-C < 1.8 mmol/L is reasonable for most patients with CCS. A target of < 1.4 mmol/L is recommended for patients with a history of ACS, recurrent events or those with additional CV risk factors.
- Targets may not be feasible when LDL-C baseline levels are very high, even with the use of PCSK9i. For these patients, treatment should aim for a $\geq 50\%$ reduction from baseline as starting goal.

(4) Patients with CKD (Refer to [NUP Chronic Kidney Disease CPG](#) for further information)

- Use an ACE inhibitor or ARB titrated to maximum tolerated dose as needed, with addition of an SGLT2 inhibitor for patients with CKD and persistent albuminuria, regardless of diabetes status

(5) Patients with chronic heart failure (Refer to [NUP Heart Failure CPG](#) for further information)

- Pharmacological treatment for patients with CHF is influenced by several factors, including the type of heart failure, the need for symptomatic management (such as using diuretics for fluid overload), and interventions aimed at improving prognosis
- Patient specific considerations, such as comorbidities, medication tolerability, and adherence, also play a crucial role in determining the most appropriate management strategy
- For patients with reduced left ventricular ejection fraction ($\leq 40\%$), an effective combination of medications typically includes SGLT2 inhibitors, angiotensin receptor-neprilysin inhibitors, mineralocorticoid receptor antagonists, and beta blockers

(D) Oral anticoagulant therapy (Refer to [NUP Non-Valvular Atrial Fibrillation CPG](#) for further information)

- For patients with new-onset AF with no recent stent (within the past 12 months), consider oral anticoagulant monotherapy based on modified CHA₂ DS₂ VASc score and patient factors such as comorbidities and bleeding risk (OAC therapy alone is sufficient)
- For patients with new-onset AF and a stent within the past 12 months, refer to cardiologist to reassess and optimise the current antithrombotic therapy

Non-Pharmacological Treatment

- Lifestyle interventions can significantly improve the overall health status of patients with CCS, and their benefits are well established in international guidelines and the literature
- These interventions warrant a multidisciplinary approach, and include increased physical activity, dietary changes, and **smoking cessation**
- The goals are to enhance patients' knowledge, self-care abilities, and health-related quality of life

Lifestyle Advice for Patients with CCS

BMI and Weight Management	• Encourage patients to achieve and maintain a healthy weight (BMI < 23 kg/m²) • Advise patients to lose weight (if required) by adhering to the recommended energy intake and increasing physical activity (± pharmacological management)
Psychosocial Aspects	• Advise patients to avoid situations that can induce psychosocial stress • Treat depression and anxiety through psychological or pharmacological interventions
Sexual Activity	• Inform patients that sexual activity is associated with low risk of CV complications if their CCS is stable and asymptomatic at low-to-moderate activity levels • Advise male patients that PDE-5 inhibitors are generally safe, but not to be taken in combination with nitrate medications because of risk of severe hypotension (Refer to NUP Erectile Dysfunction CPG for further information)
Patient Education	• Educate patients about the condition, the importance of both pharmacological and non-pharmacological interventions (including the role of physical activities and exercise), and self-care • Remind patients about the importance of adhering to medications and lifestyle interventions

Recommending Exercise for Patients with CCS

- Regular physical activity, including systematic exercise, is an important component of therapy for CCS and is associated with reduced cardiovascular and all-cause mortality.
- All individuals with established (long-standing) CCS should be encouraged to perform the minimal physical activities recommendations for general and CV health. This applies to individuals with stable angina, asymptomatic and symptomatic individuals stabilized < 1 year after ACS, or individuals with recent revascularization, and asymptomatic and symptomatic individuals > 1 year after initial diagnosis or revascularization.

- Asymptomatic individuals with long-standing CCS who intend to participate in intensive exercise or competitive sports should be referred to the cardiologist for further assessment as they may require exercise stress testing or functional imaging and echocardiography.
- Every patient should be thoroughly informed that development of symptoms during exercise should prompt reassessment.
- Individuals taking dual anti-platelet agents should avoid sports with bodily collision, especially when they are combined with oral anticoagulants, due to the risk of haemorrhage.
- Exercise is contraindicated in patients with the following clinical conditions, and further assessments or advice from the specialist / cardiologist may be needed:
 - Patients with unstable conditions (e.g. uncontrolled hypertension or DM)
 - Anginal symptoms
 - High-grade arrhythmias (e.g. ventricular fibrillation)
 - Decompensated heart failure
 - Severe aortic dilatation
 - Active thromboembolic disease

Follow Up and Monitoring

- Schedule regular follow-up visits for all patients with CCS to monitor symptoms, assess medication adherence, and adjust treatment plans as needed.
- During follow-up visits:
 - Assess overall CV risk factors especially in patients with comorbid conditions (e.g. dyslipidaemia, T2DM)
 - Review any reported symptoms on exertion and at rest, and their impact on daily activities
 - Assess adherence to non-pharmacological advice (e.g. physical activities) and medications
 - Remind patients to keep up to date with vaccination against influenza, pneumococcal disease, other widespread infections (e.g. COVID-19)
 - If required, refer to a tertiary centre or specialist for further tests or collaborative care, for example patients with heart failure or poorly controlled angina

Role of Health Team Members

1. Family Physician

- To provide management of CCS, including health promotion and prevention / detection / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient
- Introduce and enrol patient onto Teleconsult
- Perform general screening (i.e., fall risk, social economics, smoking & drinking history)
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza and pneumococcal) and cancer screenings (FIT kit, mammogram, PAP smear / HPV)
- Teach home blood pressure monitoring to patient and validate BP set
- Recruitment into PTEC - HT

3. Care Manager

- Patient education on CCS, BP targets, LDL targets and glucose targets, and lifestyle measures
- Teach home BP monitoring
- Assess for treatment adherence
- Alternate CM visit with doctor visit for patients with stable disease and when comorbid conditions are under control
- Recruitment into PTEC - HT if applicable

4. Advanced Practice Nurse (APN)

- Manage patients with diabetes, hypertension and dyslipidaemia within scope of practice
- Titrate medications for hypertension, diabetes and hyperlipidaemia
- Referral to SOC can be made by APNs

5. Clinical Pharmacist

- Titrate medications for hypertension, diabetes and hyperlipidaemia
- For patients who need drug optimisation or closer monitoring due to increased risk for drug interactions, polypharmacy, lack of drug-related knowledge or drug non-adherence
- Optimise cardiac related medications for secondary prevention of ACS

6. Dietitian

- Patient education on DASH diet
- Patient education on weight management
- Patient who needs more dietary / nutritional support / advise on food choices

7. Psychologist

- Psychological and behavioural interventions to manage psychological stress, improve disease management and quality of life.
- Psychological assessment and intervention to manage co-occurring psychological problems (depression, anxiety disorders etc.)
- Psychological support for carers experiencing caregiver stress

8. Medical Social Worker

- Conduct biopsychosocial assessment
- Provide intervention for social assistance including application of schemes and services, and referral for community resources and caregiver support to support with living with their chronic disease

9. Financial Counsellor

- Financial counselling advice with patients raising difficulties in managing their medical bills
- Advise on MediSave utilisation for chronic disease polyclinic bills
- Application on government endowment fund- MediFund and Medication Assistance Fund for subsidies on high-cost drugs
- Application on Institutional Medical Fund-Remission (if applicable for non-Singaporeans)

10. Pharmacist

- Smoking cessation clinic
- Detect, prevent and address drug-drug / drug-disease interactions
- Perform medication reconciliation

ASSESSMENT OF ACUTE EXACERBATION OF CHEST PAIN IN PATIENTS WITH CCS

- Patients with established CCS can present with signs and symptoms of acute chest pain (angina). In these circumstances, establishing the origin of the chest pain can be challenging, requiring detailed clinical history and assessment of concomitant risk factors.
- The primary aim of the assessment is to determine whether the chest pain is of cardiac origin, which may indicate an emergency
- 'Typical' chest pain is more likely to be of cardiac origin and presents with all three of the following features:
 - Constricting or compressive discomfort in the front of the chest, radiating to the neck, shoulders, jaw, or arms
 - Precipitated by physical exertion
 - Relieved by rest or short-acting glyceryl trinitrate within approximately 5 minutes
- When only (any) two of the above are present, chest pain is less likely to be of cardiac origin
- When chest pain is suspected to be of cardiac origin, a resting 12-lead ECG test can offer baseline assessment at presentation. All patients with CCS with chest pain of suspected cardiac origin should be offered referral either to the ED for further assessment or for urgent review by cardiologist, depending on the clinical picture.
- For patients with CCS and non-cardiac chest pain, other possible causes of chest pain should be considered, and managed accordingly

FITNESS CERTIFICATION

1. Class 1, 2 and 3 Licences
 - Angina - Not fit until angina is satisfactorily controlled
 - Myocardial infarction, coronary artery bypass graft, unstable angina - Not fit for at least 1 month after episode.
 - Coronary angioplasty - At least 1 week off driving; resume driving if recovery satisfactory
2. Class 4, 5 and Vocational Licences
 - Completion of exercise stress test required - will require cardiologist certification

REFERRAL BACK TO CARDIOLOGY SOC

Conditions that warrant urgent review back at SOC:

1. Angina/ chest pain
 - New onset of chest pain suspected to be ischaemic of origin
 - Exacerbation of stable angina
 - Recurrence of angina
2. New onset heart failure, or exacerbation of heart failure
3. New onset atrial fibrillation that is highly symptomatic or with poor rate control (heart rates < 40bpm or > 110bpm at rest)
4. Syncope
5. Problems with employment, life insurance, or unacceptable interference with lifestyle
6. Patient wishes for referral back to cardio (5 and 6 can be routine referral)

RECOMMENDED CARE COMPONENTS

Recommended Care Components[#]	Minimum Frequency*	Remarks
Blood Pressure Measurement	Twice a year	
Weight and BMI Assessment	Twice a year	Keep < 23kg/m ² (for non-Asian population, keep BMI < 25 kg/m ²)
Lipid Profile	Annually	Target LDL < 1.8 mmol/L and < 1.4 mmol/L for patients with history of ACS
Smoking Assessment	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provision of smoking cessation counselling
Diabetes Screening	Annually or once every three years, as clinically indicated	Screening should be carried out every three years for those with normal glucose tolerance, and annually for those with impaired fasting glycaemia (IFG) or impaired glucose tolerance (IGT)
Kidney Function Monitoring	Annually	More frequently for patients on ACE inhibitors. Urine Albumin-Creatinine (uACR) annually or more frequently if abnormal
Influenza Vaccination	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS)
Pneumococcal Vaccination	As recommended under NAIS	As recommended under NAIS
Shingles Vaccination (Recombinant herpes zoster vaccine)	2 doses at an interval of 2 to 6 months	As recommended under the NAIS and NCIS (patients aged 60 years or older)

REFERENCES

1. 2025 ACE Clinical Guidance - Management of chronic coronary syndrome
2. 2020 ESC Guidelines on sports cardiology and exercise in patients with cardiovascular disease
3. Chronic Disease Management Programme – Handbook for Healthcare Professionals. Ministry of Health Singapore 2022